BERKELEY BioMechanics

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3D-Bioprinted Cartilage Implants to Improve Treatment Options for Osteoarthritis Patients

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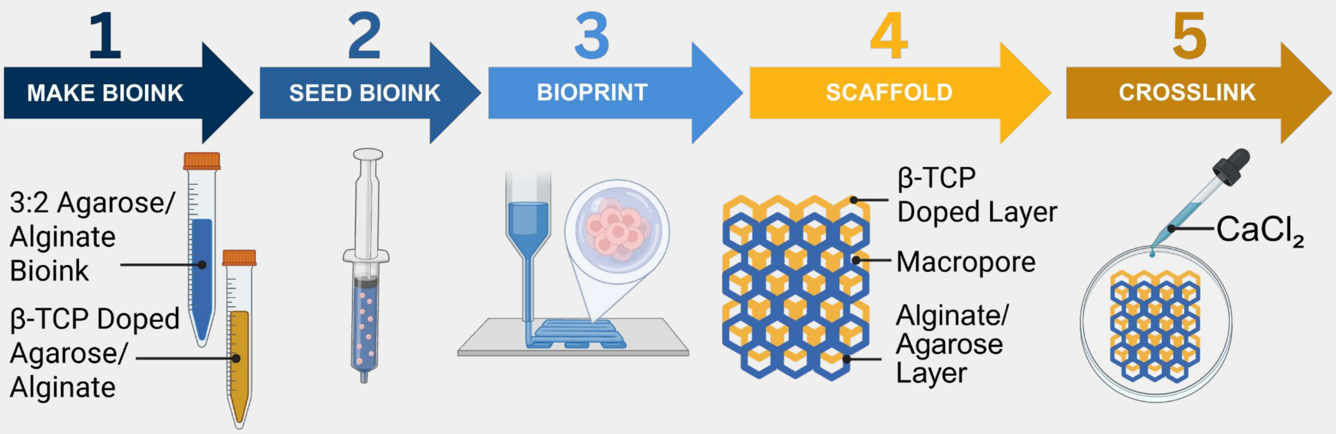
32.5 million Americans suffer from cartilage damage due to osteoarthritis. Total knee replacement is invasive and does not restore full function, making it less ideal for younger, active patients.

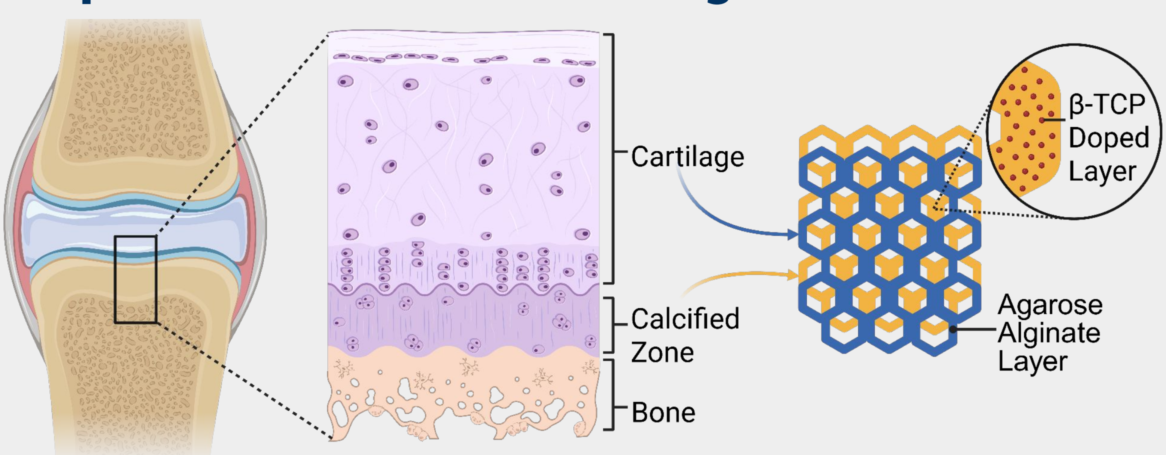
This project develops biomimetic **3D-bioprinted macroporous** scaffolds to improve nutrient diffusion, cell viability, and osteointegration, enabling longer-lasting cartilage repair and delaying joint replacement.

Bioprinting Enables Controlled Scaffold Architecture

Bioprinting Strategy Enables Multilayer Cartilage Scaffolds



Novel Implants Mimic Native Cartilage Architecture



Layered scaffold mimics native cartilage: β-TCP bioink forms the mineralized calcified layer aiding osteointegration, while β-TCP–free hydrogel forms the articular cartilage layer, improving implant mechanics and integration with surrounding tissue.

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